7.1 Normal Vector and Cartesian Equation

Definition and results

A normal non-zero vector at a straight line is orthogonal to any leading vector of this Right. $\vec{n} = (a; b) \neq (0; 0)$ is normal and $A(x_A; y_A)$ belongs to d, then d a for equation $a(x - x_A) + b(y - y_A) = 0$, or $ax + by + c = 0$. A possible guiding vector is ($-b; a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM} \cdot \vec{n} = 0$. In an orthonormal coordinate system, develop this product gives the equation. Conversely the set thus defined is the line passing through A orthogonal to n.

Example solved. By $A(1; 2)$ and normal $(2; -3) : 2(x - 1) - 3(y - 2) = 0$, therefore $2x - 3y + 4 = 0$. The vector (3; 2) directs the right: produces $6 - 6 = 0$.

Point of vigilance

$(a; b)$ is normal, and not Director in general. ($a; b)$ must not be void.

Immediate application. Equation of the right by (2; $-1)$ normal (3; 4).

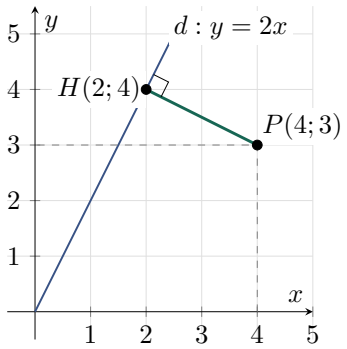
7.2 Orthogonal project and distance

Definition and results

Projected H of P over d is characterized by $H \in d$ and $\overrightarrow{PH}$ colinear to a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a; b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. Distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a deduced complement, not an automatism Isolated.

Status and justification. Replace $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ in the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; H is obtained and $PH = |\lambda| \sqrt{a^2 + b^2}$.

Example solved. For $P(3; 4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3 + 4 - 2}{2} = \dfrac{5}{2}$, therefore $H(\dfrac{1}{2}; \dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



The PH segment is perpendicular to d. The two axes have the same scale; the coordinates of P and H read directly.

Point of vigilance

The distance is positive or zero: keep the absolute value. Two checks, belonging and perpendicularity, secure the project.

Immediate application. Project $P(4; 1)$ on $d : x = 2$.

7.3 Equation of a circle

Definition and results

The center circle $I(\alpha; \beta)$ and radius $r > 0$ a for equation ($x - \alpha)^2 + (y - \beta)^2 = r^2$. One equation $x^2 + y^2 + ux + vy + w = 0$ is processed by completing the two squares. square obtained is negative, the whole is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Both members being positive or null, one can raise squared and use the orthonormal distance formula.

Example solved. $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 4$. Centre $(2; -3)$, radius 2. The signs of the centre coordinates are opposite to those in parentheses.

Point of vigilance

A quadratic equation in x and y is not always a circle: check the coefficients and the real beam.

Immediate application. Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$.

7.4 Solve a configuration

Definition and results

An intersection is obtained by simultaneously solving the equations.
straight parallel to an axis, substitute the constant coordinate gives an equation of the second degree. For a mediator, write equality of distances squared; quadratic terms eliminate.

Status and justification. Equivalencies are based on belonging to both sets.
$MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Example solved. The circle $x^2 + y^2$ = 25 and the right x = 3 cut into (3; 4) and (3; -4. The mediator of A(0; 0) and B(4; 2) check $x^2 + y^2 = (x-4)^2 + (y-2)^2$, therefore $2x + y = 5$.

Point of vigilance

An intersection is not the union of solutions sets. Check the points in both equations.

Immediate application. Intersect ($x - 1)^2 + (y + 2)^2$ = 9 with $y = 1$.